

Mingyu Wang

New Taipei City, Taiwan

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Data Analyst | Financial & Risk Analytics | Python / SQL / R

PROFESSIONAL SUMMARY

Data Analyst with a background in Information Management and completed MSc-level training in Financial Technology with Data Science at the University of Bristol, with final results pending. Experienced in Python, SQL, R, statistical modelling, customer analytics, financial and transaction data, relational data modelling, and reproducible analytical workflows. Skilled at turning complex or loosely defined datasets into structured, interpretable analysis to support business, risk, and product decision-making.

CORE SKILLS

Analytics and Programming: Python, SQL, R, pandas, NumPy, Polars, scikit-learn, exploratory data analysis, statistical modelling, feature engineering, data visualisation, validation

Financial and Risk Analytics: financial market data, market microstructure, order-book analytics, customer behaviour analytics, risk indicators, segmentation, high-frequency data, transaction data

Data Systems and Cloud: relational data modelling, PostgreSQL / Supabase, data cleaning, data validation, reproducible pipelines, AWS (EC2, S3, SQS, DynamoDB, Auto Scaling, CloudWatch, IAM)

Development and Reporting: Git, GitHub, Jupyter Notebook, Tableau, R Markdown, Next.js, TypeScript, Django, Solidity, Hardhat, Wagmi/viem

EDUCATION

University of Bristol

MSc Financial Technology with Data Science

Bristol, United Kingdom

Sep 2025 – Sep 2026

- Global rankings: TIME 2026 #35; QS 2027 #57; Times Higher Education 2026 =#80.
- Completed all taught modules, MSc Individual Project, and viva examination; final project mark, programme results, and formal degree award pending, with final results expected in November 2026.

National Kaohsiung University of Science and Technology

Bachelor of Science in Information Management

Kaohsiung, Taiwan

Sep 2020 – Jun 2024

- Studied data science, database systems, software engineering, big data analytics, social network analysis, and information systems, with projects using Python, R, SQL, data processing, visualisation, databases, and web application development.

SELECTED ANALYTICS PROJECTS

High-Frequency Liquidity Shock Detection and Order-Book Resiliency | MSc Individual Project

- Analysed one-second Databento XNAS.ITCH MBP-10 order-book data for 10 Nasdaq-listed securities across 37 trading sessions, identifying 34,589 spread- and depth-family shock records.
- Built an outcome-blind event-detection and recovery framework using historical time-of-day median/MAD benchmarks, first-passage T50/T90 measures, censor-aware analysis, and pipeline validation.
- Applied Kaplan–Meier estimation and security-stratified log-rank testing; pooled reached-only median T90 was 3 seconds for quoted spread versus 10 seconds for displayed depth.
- Validated 34,589 event records across 3.73 million event-second recovery rows with zero dropped or duplicated events and independently checked T50/T90 calculations.

MindPass FinTech Escrow DApp | FinTech / Relational Data / Smart Contracts

- Developed a four-person hybrid FinTech MVP combining smart-contract escrow with a PostgreSQL / Supabase relational data layer for booking, funding, subsidy, refund, and settlement workflows.
- Used relational tables and SQL-backed application logic to connect users, sessions, funding records, subsidy state, transaction records, and settlement-related workflow states.
- Integrated on-chain events with PostgreSQL / Supabase application data so booking, funding, refund, and settlement information could be reflected consistently across the application workflow.
- Led technical implementation, integration, and Sepolia validation across workflows covering split funding, provider decisions, check-in and no-show handling, refunds, claimable balances, and withdrawals using Next.js, TypeScript, Solidity, and Hardhat.

Customer Behaviour Risk Analytics & Segmentation | R / Customer Analytics

- Built a customer risk analytics case study using trading data from 1,000 customers, translating transaction-level activity into interpretable customer-level behavioural indicators.
- Engineered signals for high-exposure trades, FOMO-like buying, market-order reliance, and repeated deposits under low-balance conditions.
- Identified the 200 most active customers and selected a 40-customer target cohort using behavioural risk indicators for descriptive risk segmentation and prioritised monitoring.
- Implemented the analysis in R using a reproducible R Markdown workflow, including feature engineering, aggregation, cohort selection, visualisation, and documented limitations.

Robust Optimisation and Predictive Modelling Pipeline | R / Statistical Modelling

- Built reproducible R-based modelling workflows covering exploratory analysis, model comparison, validation, holdout evaluation, and interpretation.
- Evaluated five Differential Evolution configurations across 26 benchmark problems using single-problem and multi-problem blocked designs to assess robustness and decision stability.
- Built a leakage-aware predictive modelling workflow using group-aware data splitting, grouped 5-fold cross-validation, and held-out evaluation.

LEADERSHIP EXPERIENCE

Chair, Student Judicial Council

National Kaohsiung University of Science and Technology

- Led structured review and discussion of student affairs, coordinating meetings, documentation, and communication across student governance stakeholders.

Deputy Speaker, Student Parliament

National Kaohsiung University of Science and Technology

- Supported parliamentary discussions and communication between student representatives and university stakeholders, developing cross-functional coordination and stakeholder communication skills.

LANGUAGES AND ADDITIONAL INFORMATION

- **Languages:** Mandarin Chinese: Native; English: Professional working proficiency; MSc programme completed in English.
- **Career Interests:** Data Analytics, Business Analytics, Financial Data Analytics, Risk Analytics, Customer Analytics, Product Analytics, Market Data, and FinTech.
- **Location and Availability:** Based in New Taipei City; open to opportunities across Taiwan and available for full-time roles from October 2026.
- **Work Authorisation:** Eligible to work in Taiwan.